Akshith Macharla

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Education

California State University, Chico — Chico, CA
Master of Science in Computer Science – GPA 3.96/4.00 (degree courses) — *Expected Dec 2026*

Chaitanya Bharathi Institute of Technology — Hyderabad, India
Bachelor of Engineering in Electrical and Electronics Engineering — *2017 – 2021*

Technical Skills

- **Languages:** Python, C++, CUDA, Java, SQL | **Infrastructure:** Linux, Docker, AWS, Git, pytest
- **LLM Inference:** vLLM, FlashInfer, PyTorch, prefill/decode, speculative decoding, paged KV cache, FP8/NVFP4/INT8
- **Performance Analysis:** roofline analysis, Nsight Systems, PyTorch Profiler, TTFT/TPOT and p50/p95 latency, NCCL bandwidth
- **Agents:** LLM tool-calling agents (OpenAI API), agent guardrails and output verification, coding agents (Claude Code, Codex)
- **Accelerators & Distributed:** CUDA, CUTLASS, Triton, AWS Neuron (NKI), NCCL, PyTorch Distributed; A100, H100, Inferentia2

Open-Source Contributions

AWS Labs keys_values – LLM inference library: speculative decoding, KV cache | *Python, PyTorch, CUDA* — *1 merged, 2 open*
- **Speculative decoding (PR #164, open):** draft-then-verify generation with exact rejection sampling, so outputs keep the target model's distribution; 1.12× decode throughput with a Qwen3-14B target and 0.6B draft at 73.6% acceptance (RTX 6000 Ada).
- **Tentative KV-cache updates (PR #168, open):** on the maintainer's buffer-switching design, keeps accepted draft tokens and rolls back the rest for dense, sliding-window, and 8/4-bit quantized caches; bit-exact restoration, 2,000+ GPU test cases.
- **Per-sequence KV-cache quantization fix (PR #155, merged):** fixed 4/8-bit bitsandbytes blocks spanning sequences in a batch, where perturbing one sequence changed 3,052 of 49,152 values in the others; validated on an A40 across 1,255 test cases.

AWS Neuron nki-samples – NKI kernel port to Neuron SDK 2.32 | *Python, NKI* — *Merged | PR #138*
- Ported a max-pooling kernel to NKI 0.6.0: HBM-to-SBUF DMA, on-chip max reduction, 128-partition tiles; exact on Inferentia2.

PyTorch ExecuTorch - quantized attention lowering for Arm Cortex-M | *Python, FX, PT2E* — *Merged | PR #22892*
- Wrote an SDPA decomposition pass that applies the full attention scale once after the QK matmul, not as $d^{-1/4}$ on both Q and K, so int8 scales stay proportional and fold into requantization; added end-to-end SDPA tests.

llm-compressor (vLLM project) – fused-layer quantization mappings | *Python, PyTorch, NVFP4* — *Merged | PR #3224*
- Unified fused-layer mappings (QKV, gate/up, MLA) into one resolver so packed layers share a scale; NVFP4 mismatches 8/8→0/8.

NVIDIA CUTLASS – FP8 blockwise GEMM split-K correctness fix | *C++, CUDA* — *Open | PR #3615*
- Fixed split-K blockwise-scaled GEMM silently applying slice 0's scale factors to every K-slice (output 1,280 instead of 10,880 while every status check returned `kSuccess`) by threading the K-tile offset through four mainloop layers to the scale lookup; added tests.

Projects

LLM Inference Roofline & Benchmark Agent | *Python, vLLM, OpenAI API* *github.com/amacharla15/transformer-ddp-benchmarks*
- Built a roofline model of Qwen3-8B inference (bytes vs. FLOPs per forward pass) predicting TTFT, TPOT, and the decode ridge point; on an H100 with vLLM across 10 configurations, decode reached 69–81% and prefill 36–61% of the roofline.
- Added an OpenAI tool-calling agent that picks configurations to benchmark from the predictions under code-enforced guards; it found the largest batch under a 20 ms TPOT target (112) in 4 GPU runs, where the roofline alone accepted batch 128 (20.2 ms).

Multi-GPU Serving & Collectives Benchmarks | *vLLM, NCCL, PyTorch Distributed* *github.com/amacharla15/pytorch*
- Benchmarked vLLM TP=2 serving (TTFT, TPOT); NCCL all-reduce 172.6 GB/s and broadcast 252.9 GB/s on two NVLink A100s.

CUDA Transformer Kernels | *CUDA, C++, PyTorch, Nsight Systems* *github.com/amacharla15/CUDAkernels*
- Hand-wrote tiled GEMM, multi-head attention, and FlashAttention-style attention with online softmax, checked against PyTorch.

LLM Inference Kernels | *CUDA, C++* *github.com/amacharla15/fused-llm-inference-kernels-in-cuda*
- CUDA kernels for a Llama-style FFN block: RoPE, causal softmax, SwiGLU, fused add + RMSNorm, fused linear + GELU.

Experience

Cognizant Technology Solutions — Hyderabad, India
Programmer Analyst / Associate Software Engineer — *Dec 2021 – Aug 2024*
- Built and maintained Java/Spring Boot and PostgreSQL services on AWS for policy lookup, claims, and audit history (10,000+ users).
- On-call with Dynatrace; triaged and root-caused 100+ production incidents (API failures, data inconsistencies, regressions).
- Cut p95 API latency from ∼250 ms to ∼200 ms by tracing slow requests to redundant reads and slow Hibernate/SQL queries.
- Implemented REST endpoints with request validation and centralized exception handling; wrote JUnit/Mockito tests.

California State University, Chico — Chico, CA
Research Assistant, 3D Computer Vision — *Feb 2026 – Jun 2026*
- Built C++/OpenCV tools and Linux pipelines for RealSense RGB-D data: depth/color alignment, filtering, and road-plane fitting.